

Figure 1: Feature construction for teacher A. The 21-dimensional base representation contains joint state, robot-relative target geometry, FK target error and world target height. The gate variant appends one observable reach-active indicator before fitting-only standardisation.

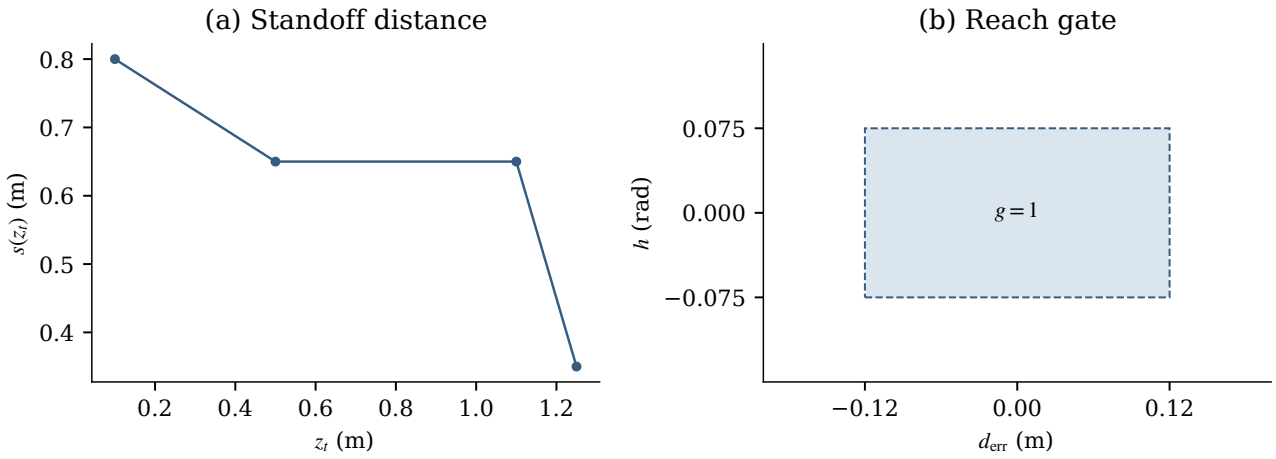


Figure 2: (a) Height-dependent standoff distance. (b) Reach gate in distance error and heading error: $g = 1$ only where $|d_{\text{err}}| < 0.12\,\text{m}$ and $|h| < 0.075\,\text{rad}$. The dashed boundary is excluded. The gate is reconstructed from current deployable features, without teacher memory.

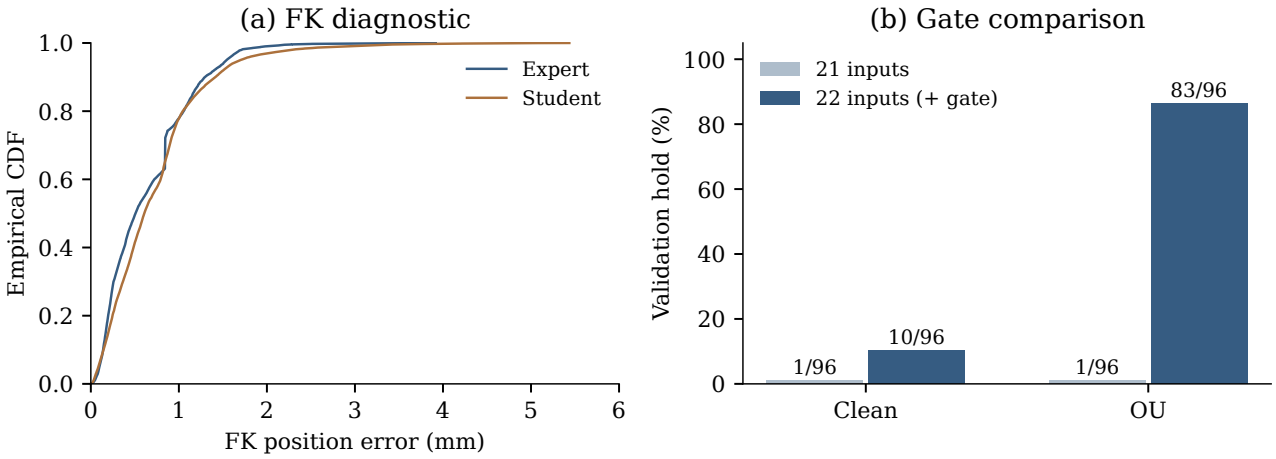


Figure 3: (a) Deployed FK error on four nominal fitting scenarios per role, sampled every 10 ms. The student is the archived 27 October DAgger round-1 student, not ablation row R1. (b) Gate comparison for the fixed v1 rows R1–R4, pooled over three training seeds on the same 32 validation scenes. Each denominator is 96 rollouts, not 96 distinct scenarios.